D.5 Proof of Proposition 5.1

Proof of Proposition 5.1. First, notice that by assuming $\sum_{i\in\mathcal{I}_2}\hat{f}_{\tau_{\text{prior}}}(X_i) \leq B$, is follows that $\pi_i^* = 1$ for all $i \in \mathcal{I}_2$ is the optimal solution which satisfies the constraint. We now turn to analyze the setup where $\sum_{i\in\mathcal{I}_2}\hat{f}_{\tau_{\text{prior}}}(X_i) > B$. We prove by induction over the sum $\sum_{i\in\mathcal{I}_2}\hat{f}_{\tau_{\text{prior}}}(X_i)$, assuming that $\hat{f}_{\tau_{\text{prior}}}(X_i) \leq M\ \forall i \in \mathcal{I}_2$.

Base case. We begin with the maximal sum: $\sum_{i\in\mathcal{I}_2}\hat{f}_{\tau_{\text{prior}}}(X_i) = |\mathcal{I}_2|M$. That is, $\hat{f}_{\tau_{\text{prior}}}(X_i) = M$ for all $i \in \mathcal{I}_2$. Then, by the optimization constraint, we get

$$M\sum_{i\in\mathcal{I}_2}\pi_i^* = B \qquad \Longrightarrow \qquad \sum_{i\in\mathcal{I}_2}\pi_i^* = \frac{B}{M},$$

and by symmetry the unique minimizer of $\sum_i 1/\pi_i^*$ with $\sum_i \pi_i^* = B/M$ is

$$\pi_i^* = \frac{B}{n\,M} \quad \forall i \in \mathcal{I}_2.$$

Inductive step. We now suppose that the claim holds for any weight vector with a sum $B-1 < S \leq M|\mathcal{I}_2|$, and show that it holds for any vector with a sum $S-1$. Given a weight vector $\{\hat{f}_{\tau_{\text{prior}}}(X_i)\}_{i\in\mathcal{I}_2}$ with a sum $\sum_{i\in\mathcal{I}_2}\hat{f}_{\tau_{\text{prior}}}(X_i) = S-1$, we compose a new vector with a sum S, as follows. In $\{\hat{f}_{\tau_{\text{prior}}}(X_i)\}_{i\in\mathcal{I}_2}$, there exists an index i_0 such that $\hat{f}_{\tau_{\text{prior}}}(X_{i_0}) \leq M-1$. The new vector with a sum S is defined by:

$$\hat{f}'_{\tau_{\text{prior}}}(X_i) = \begin{cases} \hat{f}_{\tau_{\text{prior}}}(X_i), & i \neq i_0, \\ \hat{f}_{\tau_{\text{prior}}}(X_i)+1, & i = i_0, \end{cases}$$

Notice that $\max_i \hat{f}'_{\tau_{\text{prior}}}(X_i) \leq M$ and $\sum_{i\in\mathcal{I}_2}\hat{f}'_{\tau_{\text{prior}}}(X_i) = S$. Let λ and λ' be the unique solutions of

$$U(\lambda) = B \quad \text{and} \quad U'(\lambda') = B,$$

where

$$U'(\lambda) = \sum_{i\neq i_0}\hat{f}_{\tau_{\text{prior}}}(X_i)\,\min\Big\{1, \tfrac{1}{\sqrt{n\lambda\hat{f}_{\tau_{\text{prior}}}(X_i)}}\Big\} + (\hat{f}_{\tau_{\text{prior}}}(X_{i_0})+1)\,\min\Big\{1, \tfrac{1}{\sqrt{n\lambda(\hat{f}_{\tau_{\text{prior}}}(X_{i_0})+1)}}\Big\}.$$

Since for every $\lambda > 0$ we have

$$\frac{\hat{f}_{\tau_{\text{prior}}}(X_{i_0})+1}{\sqrt{|\mathcal{I}_2|\lambda(\hat{f}_{\tau_{\text{prior}}}(X_{i_0})+1)}} = \frac{\sqrt{\hat{f}_{\tau_{\text{prior}}}(X_{i_0})+1}}{\sqrt{|\mathcal{I}_2|\lambda}} > \frac{\sqrt{\hat{f}_{\tau_{\text{prior}}}(X_{i_0})}}{\sqrt{|\mathcal{I}_2|\lambda}} = \frac{\hat{f}_{\tau_{\text{prior}}}(X_{i_0})}{\sqrt{|\mathcal{I}_2|\lambda\hat{f}_{\tau_{\text{prior}}}(X_{i_0})}},$$

and of course $\hat{f}_{\tau_{\text{prior}}}(X_{i_0})+1 > \hat{f}_{\tau_{\text{prior}}}(X_{i_0})$, we get the inequality

$$(\hat{f}_{\tau_{\text{prior}}}(X_{i_0})+1)\,\min\Big\{1, \tfrac{1}{\sqrt{|\mathcal{I}_2|\lambda(\hat{f}_{\tau_{\text{prior}}}(X_{i_0})+1)}}\Big\} > \hat{f}_{\tau_{\text{prior}}}(X_{i_0})\,\min\Big\{1, \tfrac{1}{\sqrt{|\mathcal{I}_2|\lambda\hat{f}_{\tau_{\text{prior}}}(X_{i_0})}}\Big\},$$

and thus,

$$\begin{aligned} U'(\lambda) &= \sum_{i\neq i_0}\hat{f}_{\tau_{\text{prior}}}(X_i)\,\pi_i^*(\lambda) + (\hat{f}_{\tau_{\text{prior}}}(X_{i_0})+1)\,\min\Big\{1, \tfrac{1}{\sqrt{|\mathcal{I}_2|\lambda(\hat{f}_{\tau_{\text{prior}}}(X_{i_0})+1)}}\Big\} \\ &> \sum_{i\neq i_0}\hat{f}_{\tau_{\text{prior}}}(X_i)\,\pi_i^*(\lambda) + \hat{f}_{\tau_{\text{prior}}}(X_{i_0})\,\min\Big\{1, \tfrac{1}{\sqrt{|\mathcal{I}_2|\lambda\hat{f}_{\tau_{\text{prior}}}(X_{i_0})}}\Big\} \\ &= U(\lambda) = B. \end{aligned}$$

Since U' is continuous and decreasing on $(0,\infty)$ with $\lim_{\mu\to 0^+}U'(\mu) = \sum_i \hat{f}_{\tau_{\text{prior}}}(X_i) > B$ and $\lim_{\mu\to\infty}U'(\mu) = 0$, the intermediate-value theorem guarantees a unique λ' with $U'(\lambda') = B$. Moreover, since U' is decreasing, and

$U'(\lambda) \geq B = U'(\lambda')$ we get $\lambda' \geq \lambda$. Define π_i the optimal solution of (10) with $\hat{f}_{\tau_{\text{prior}}}(X_i)$ and π'_i the optimal solution with $\hat{f}'_{\tau_{\text{prior}}}(X_i)$.

- If $i \neq i_0$, since $\hat{f}'_{\tau_{\text{prior}}}(X_i) = \hat{f}_{\tau_{\text{prior}}}(X_i)$,

$$\pi'_i = \min\Big\{1, \frac{1}{\sqrt{|\mathcal{I}_2|\,\lambda'\,\hat{f}'_{\tau_{\text{prior}}}(X_i)}}\Big\} \;\leq\; \min\Big\{1, \frac{1}{\sqrt{|\mathcal{I}_2|\,\lambda\,\hat{f}_{\tau_{\text{prior}}}(X_i)}}\Big\} = \pi_i.$$

- If $i = i_0$, then $\hat{f}'_{\tau_{\text{prior}}}(X_{i_0}) = \hat{f}_{\tau_{\text{prior}}}(X_{i_0}) + 1 > \hat{f}_{\tau_{\text{prior}}}(X_{i_0})$, so

$$\pi'_{i_0} = \min\Big\{1, \frac{1}{\sqrt{|\mathcal{I}_2|\,\lambda'\,(\hat{f}_{\tau_{\text{prior}}}(X_{i_0})+1)}}\Big\} \;\leq\; \min\Big\{1, \frac{1}{\sqrt{|\mathcal{I}_2|\,\lambda'\,\hat{f}_{\tau_{\text{prior}}}(X_{i_0})}}\Big\} \;\leq\; \min\Big\{1, \frac{1}{\sqrt{|\mathcal{I}_2|\,\lambda\,\hat{f}_{\tau_{\text{prior}}}(X_{i_0})}}\Big\} = \pi_{i_0}.$$

By the inductive assumption, $\pi'_i \geq B/(|\mathcal{I}_2|M)$ for all $i \in \mathcal{I}_2$, hence $\pi_i \geq B/(|\mathcal{I}_2|M)$ for all $i \in \mathcal{I}_2$ as well. Lastly, by the definition of the weights, we have:

$$w(\{X_j\}_{j\in\mathcal{I}_2}, i) = (\pi^*_i)^{-1} \leq \max\left(\frac{|\mathcal{I}_2|M}{B}, 1\right). \tag{12}$$

This completes the proof. □

D.6 Proofs of the Coverage Rate Guarantees

In this section, we present and prove the formal version Theorem 4.1 which builds on the proof of (Gui et al., 2024a, Theorem 3).

Theorem D.1 (General validity, formal)**.** *Fix a tolerance level $\delta \in (0,1)$ and a miscoverage level $\tau \in (0,1)$. Suppose that $\{(X_i, T_i)\}_{i=1,\ldots,n}$ and $(X_{\text{test}}, T_{\text{test}})$ are drawn i.i.d., and that the censoring times satisfy the conditional independence assumption (Assumption 2.1) and $(C_i, T_i) \perp\!\!\!\perp (C_j, T_j) | \{X_k\}_{k\in\mathcal{I}_2}$ for all $i \neq j \in \mathcal{I}_2$. We suppose that $\hat{q}_\tau(x)$ is non-decreasing and continuous in τ. Further, assume that the weights are computed using the true probabilities:*

$$w_\tau(\{X_j\}_{j\in\mathcal{I}_2}, i) = 1/\mathbb{P}\big[\hat{q}_\tau(X_i) \leq C_i \;\big|\; \{X_j\}_{j\in\mathcal{I}_2}\big]. \tag{13}$$

Above, $\hat{q}_\tau(X_i)$ can be either the estimated quantile of $T_i \mid X = X_i$ presented in Section 3, or its trimmed version $\hat{f}_\tau(X_i)$ from Section 6.1. We assume that there exists a constant $\gamma_\tau > 0$ such that the weights satisfy $w_\tau(x) \leq \gamma_\tau$ for P_X-almost all x. Consider the following estimated miscoverage rate:

$$\hat{\alpha}(\tau) = \frac{1}{|\mathcal{I}_2|} \sum_{i\in\mathcal{I}_2} w_\tau(\{X_j\}_{j\in\mathcal{I}_2}, i)\, \mathbb{I}\big\{\tilde{T}_i < \hat{q}_\tau(X_i) \leq C_i\big\} \tag{14}$$

and denote the calibrated quantile level by

$$\hat{\tau} = \sup\Big\{\tau \in \mathcal{T} : \sup_{\substack{\tau'\in\mathcal{T}\\ \tau'\leq\tau}} \hat{\alpha}(\tau') \leq \alpha\Big\}. \tag{15}$$

Above, the search space $\mathcal{T}$ is formulated as in Section 4, or using $\mathcal{T} \cap [0, \tau_{\text{prior}}]$, as in the proposed `Adaptive` *method from Section 6.1. We remark that for $\hat{\tau}$ to be well defined, we assume there exists $\tau' \in \mathcal{T}$ such that $\hat{\alpha}(\tau') \leq \alpha$. This assumption can be trivially satisfied by setting $\hat{q}_0(X_i) = 0$. Then, with probability at least $1-\delta$ over the draws of $\mathcal{D}$, the LPB $\hat{L}(x) = \hat{q}_{\hat{\tau}}(x)$ satisfies*

$$\mathbb{P}\left[T_{\text{test}} \geq \hat{L}(X_{\text{test}}) | \mathcal{D}\right] \geq 1 - \alpha - \sup_{\tau\in[0,1]} \left\{\sqrt{\frac{2\gamma_\tau^2 + 5}{|\mathcal{I}_2|} \cdot \log\left(\frac{1}{\delta}\right)}\right\}. \tag{16}$$

Proof. For ease of notation, we define the coverage gap by:

$$\Delta := \sup_{\tau\in[0,1]} \left\{\sqrt{\frac{2\gamma_\lambda^2 + 5}{|\mathcal{I}_2|} \cdot \log\left(\frac{1}{\delta}\right)}\right\}. \tag{17}$$

We define the oracle miscoverage level by:

$$\tau(\alpha+\Delta) = \sup\left\{\lambda \in [0,1] : \mathbb{P}\left(T < \hat{q}_\lambda(X) \mid \mathcal{I}_1\right) \leq \alpha + \Delta\right\}. \tag{18}$$

Observe that by assuming $1-\delta \leq \mathbb{P}(\hat{\tau} \leq \tau(\alpha+\Delta) \mid \mathcal{I}_1)$, we get that the event $\{\hat{\tau} \leq \tau(\alpha+\Delta)\}$ holds with probability at least $1-\delta$. Under the monotonicity of $\hat{q}_\tau$, and following the the left-continuity of $\mathbb{P}(T \geq \hat{q}_\tau(X) \mid \mathcal{D})$ in τ we obtain that with probability at least $1-\delta$:

$$\begin{aligned}
&\mathbb{P}(T \geq \hat{q}_{\hat{\tau}}(X) \mid \mathcal{D}) \\
\geq &\mathbb{P}(T \geq \hat{q}_{\tau(\alpha+\Delta)}(X) \mid \mathcal{D}) \\
\geq &1-\alpha-\Delta.
\end{aligned} \tag{19}$$

Now, we turn to show that $1-\delta \leq \mathbb{P}(\hat{\tau} \leq \tau(\alpha+\Delta) \mid \mathcal{I}_1)$. We begin by fixing $\varepsilon > 0$, and denoting $\lambda := \tau(\alpha+\Delta)+\varepsilon$. By the definition of $\hat{\alpha}(\tau)$, we get:

$$\begin{aligned}
&\mathbb{P}(\hat{\alpha}(\tau(\alpha+\Delta)+\varepsilon) \leq \alpha \mid \mathcal{I}_1) \\
=&\mathbb{P}(\hat{\alpha}(\lambda) \leq \alpha \mid \mathcal{I}_1) \\
=&\mathbb{P}\left(\frac{1}{|\mathcal{I}_2|}\sum_{i\in\mathcal{I}_2} w_\lambda(\{X_j\}_{j\in\mathcal{I}_2}, i)\, \mathbb{I}\{\tilde{T}_i < \hat{q}_\tau(X_i) \leq C_i\} \leq \alpha \middle| \mathcal{I}_1\right) \\
=&\mathbb{P}\left(\frac{1}{|\mathcal{I}_2|}\sum_{i\in\mathcal{I}_2} w_\lambda(\{X_j\}_{j\in\mathcal{I}_2}, i)\, \mathbb{I}\{\hat{q}_\lambda(X_i) \leq C_i\}\, \mathbb{I}\{T_i < \hat{q}_\lambda(X_i)\} - \alpha \leq 0 \middle| \mathcal{I}_1\right)
\end{aligned} \tag{20}$$

Next, we apply Markov's inequality for any $t > 0$, and obtain:

$$(20) \leq \mathbb{E}\left(\exp\left(\alpha - \frac{t}{|\mathcal{I}_2|}\sum_{i\in\mathcal{I}_2} w_\lambda(\{X_j\}_{j\in\mathcal{I}_2}, i)\, \mathbb{I}\{\hat{q}_\lambda(X_i) \leq C_i\}\, \mathbb{I}\{T_i < \hat{q}_\lambda(X_i)\}\right) \middle| \mathcal{I}_1\right) \tag{21}$$

By conditioning on $\{(X_i, T_i)\}_{i\in\mathcal{I}_2}$ and following the $\frac{1}{4}$ sub-gaussianity of $w_\lambda(\{X_j\}_{j\in\mathcal{I}_2}, i)^{-1} - \mathbb{I}\{\hat{q}_\lambda(X_i) \leq C_i\}$, the conditionally independent censoring assumption (Assumption 2.1) and the bounded weights, we have:

$$\begin{aligned}
&\mathbb{E}\left(\exp\left(\frac{t}{|\mathcal{I}_2|}\cdot\sum_{i\in\mathcal{I}_2} w_\lambda(\{X_j\}_{j\in\mathcal{I}_2}, i)\mathbb{I}\{T_i < \hat{q}_\lambda(X_i)\}\left(w_\lambda(\{X_j\}_{j\in\mathcal{I}_2}, i)^{-1} - \mathbb{I}\{\hat{q}_\lambda(X_i) \leq C_i\}\right)\right) \middle| \{(X_i,T_i)\}_{i\in\mathcal{I}_2}, \mathcal{I}_1\right) \\
\leq &\exp\left(\frac{t^2}{8|\mathcal{I}_2|^2}\cdot\sum_{i\in\mathcal{I}_2} w_\lambda(\{X_j\}_{j\in\mathcal{I}_2}, i)^2\mathbb{I}\{T_i < \hat{q}_\lambda(X_i)\}^2\right) \\
\leq &\exp\left(\frac{t^2}{8|\mathcal{I}_2|^2}\cdot\sum_{i\in\mathcal{I}_2} w_\lambda(\{X_j\}_{j\in\mathcal{I}_2}, i)^2\right) \\
\leq &\exp\left(\frac{\gamma_\lambda^2 t^2}{8|\mathcal{I}_2|}\right).
\end{aligned} \tag{22}$$

Therefore,

$$\begin{aligned}
(21) &\leq \exp\left(\frac{\gamma_\lambda^2 t^2}{8|\mathcal{I}_2|}\right)\cdot\mathbb{E}\left(\exp\left(\alpha - \frac{t}{|\mathcal{I}_2|}\sum_{i\in\mathcal{I}_2} w_\lambda(\{X_j\}_{j\in\mathcal{I}_2}, i)\, w_\lambda(\{X_j\}_{j\in\mathcal{I}_2}, i)^{-1}\, \mathbb{I}\{T_i < \hat{q}_\lambda(X_i)\}\right) \middle| \mathcal{I}_1\right) \\
&= \exp\left(\frac{\gamma_\lambda^2 t^2}{8|\mathcal{I}_2|}\right)\cdot\mathbb{E}\left(\exp\left(\alpha - \frac{t}{|\mathcal{I}_2|}\sum_{i\in\mathcal{I}_2} \mathbb{I}\{T_i < \hat{q}_\lambda(X_i)\}\right) \middle| \mathcal{I}_1\right)
\end{aligned} \tag{23}$$

Further, defining $p_\lambda(\{X_j\}_{j\in\mathcal{I}_2}, i) := \mathbb{P}(T_i < \hat{q}_\lambda(X_i) \mid \{X_j\}_{j\in\mathcal{I}_2}, \mathcal{I}_1)$, we condition on $\{X_j\}_{j\in\mathcal{I}_2}$ and get, using the

$\frac{1}{4}$ subgaussianity of $p_\lambda(\{X_j\}_{j\in\mathcal{I}_2}, i) - \mathbb{I}\{T_i < \hat{q}_\lambda(X_i)\}$:

$$\begin{aligned}
&\mathbb{E}\left(\exp\left(p_\lambda(\{X_j\}_{j\in\mathcal{I}_2}, i)) - \frac{t}{|\mathcal{I}_2|}\sum_{i\in\mathcal{I}_2}(\mathbb{I}\{T_i < \hat{q}_\lambda(X_i)\}\right)\Bigg|\{X_j\}_{j\in\mathcal{I}_2}, \mathcal{I}_1\right)\\
&\leq \exp\left(\frac{t^2}{8|\mathcal{I}_2|^2}\cdot|\mathcal{I}_2|\right)\\
&= \exp\left(\frac{t^2}{8|\mathcal{I}_2|}\right)
\end{aligned} \tag{24}$$

We plug this into (23) to obtain:

$$(23) \leq \exp\left(\frac{(\gamma_\lambda^2+1)t^2}{8|\mathcal{I}_2|}\right)\mathbb{E}\left(\exp\left(\frac{t}{|\mathcal{I}_2|}\cdot\sum_{i\in\mathcal{I}_2}(\alpha - p_\lambda(X_i))\right)\Bigg|\mathcal{I}_1\right) \tag{25}$$

We now apply the Cauchy-Schwarz inequality:

$$\begin{aligned}
&\mathbb{E}\left(\exp\left(\frac{t}{|\mathcal{I}_2|}\cdot\sum_{i\in\mathcal{I}_2}(\alpha - p_\lambda(X_i))\right)\Bigg|\mathcal{I}_1\right)\\
\leq&\mathbb{E}\left(\exp\left(\frac{2t}{|\mathcal{I}_2|}\cdot\sum_{i\in\mathcal{I}_2}(\alpha - p_\lambda(X_i))\right)\Bigg|\mathcal{I}_1\right)^{1/2}
\end{aligned} \tag{26}$$

By the definition of $\tau(\alpha+\Delta)$ we have:

$$\mathbb{P}(T < \hat{q}_\lambda(X) \mid \mathcal{I}_1) \geq \alpha + \Delta. \tag{27}$$

By plugging it into the above, we get:

$$\begin{aligned}
(26) =& \mathbb{E}\left(\exp\left(\frac{2t}{|\mathcal{I}_2|}\cdot\sum_{i\in\mathcal{I}_2}(\alpha - p_\lambda(X_i))\right)\Bigg|\mathcal{I}_1\right)\\
\leq&\mathbb{E}\left(\exp\left(\frac{2t}{|\mathcal{I}_2|}\cdot\sum_{i\in\mathcal{I}_2}(\mathbb{P}(T < \hat{q}_\lambda(X) \mid \mathcal{I}_1) - \Delta - p_\lambda(X_i))\right)\Bigg|\mathcal{I}_1\right)\\
=&\exp(-2t\Delta)\mathbb{E}\left(\exp\left(\frac{2t}{|\mathcal{I}_2|}\cdot\sum_{i\in\mathcal{I}_2}(\mathbb{P}(T < \hat{q}_\lambda(X) \mid \mathcal{I}_1) - p_\lambda(X_i))\right)\Bigg|\mathcal{I}_1\right)\\
\leq&\exp\left(\frac{t^2}{2|\mathcal{I}_2|} - 2t\Delta\right).
\end{aligned} \tag{28}$$

Above, we used the $\frac{1}{4}$-sub-gaussianity of $\mathbb{P}(T < \hat{q}_\lambda(X) \mid \mathcal{I}_1) - p_\lambda(X_i)$. By combining it all, we obtain:

$$\begin{aligned}
(20) \leq& \exp\left(\frac{t^2}{2|\mathcal{I}_2|} - 2t\Delta + \frac{(\gamma_\lambda^2+1)t^2}{8|\mathcal{I}_2|} + \frac{\gamma_\lambda^2 t^2}{8|\mathcal{I}_2|}\right)\\
=& \exp\left(\frac{(2\gamma_\lambda^2+5)t^2}{8|\mathcal{I}_2|} - 2t\Delta\right)
\end{aligned} \tag{29}$$

We define

$$t = \frac{(8+2\sqrt{14})|\mathcal{I}_2|\Delta}{2\gamma_\lambda^2+5} \tag{30}$$